

WAN Authentication Protocols

Task 1: Draw a simple network diagram showing two routers connected over a WAN link and label where authentication is needed to prevent unauthorized access.

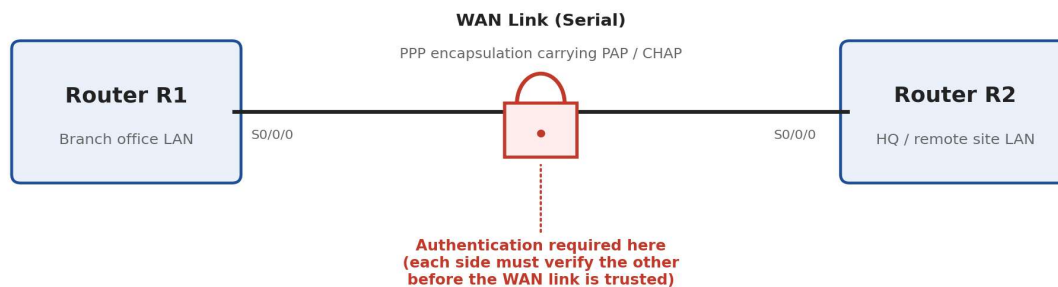


Fig. 1 — Two routers over a serial WAN link; authentication happens during PPP link negotiation, before either side trusts traffic from the other

Authentication is needed at the WAN link itself — specifically during the PPP negotiation phase on each router's serial interface (S0/0/0 in the diagram) — because this is the point where an untrusted external circuit (leased line/provider network) meets the router. Without authentication here, anyone who gains physical or logical access to that link, or spoofs the far-end router, could inject traffic into either site's LAN.

Task 2: Configure a sample HDLC WAN connection between two routers using Packet Tracer or any network simulator, and explain why HDLC does not provide authentication.

```
Router0
Router>
Router#show int
Router#show interfaces se
Router#show interfaces serial 0/0/0
Serial0/0/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.0.0.1/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations 0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
  5 minute input rate 10 bits/sec, 0 packets/sec
  5 minute output rate 13 bits/sec, 0 packets/sec
    3 packets input, 384 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    4 packets output, 512 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions
  DCD=up DSR=up DTR=up RTS=up CTS=up

Router#

Router1
Router>
Router#show int
Router#show interfaces se
Router#show interfaces serial 0/0/0
Serial0/0/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.0.0.2/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations 0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
  5 minute input rate 13 bits/sec, 0 packets/sec
  5 minute output rate 10 bits/sec, 0 packets/sec
    4 packets input, 512 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    3 packets output, 384 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions
  DCD=up DSR=up DTR=up RTS=up CTS=up

Router#
```

Router R1 (DCE end — provides clocking):

```
R1> enable
R1# configure terminal
R1(config)# interface serial 0/0/0
R1(config-if)# encapsulation hdlc
R1(config-if)# clock rate 64000
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# no shutdown
```

Router R2 (DTE end):

```
R2> enable
R2# configure terminal
R2(config)# interface serial 0/0/0
R2(config-if)# encapsulation hdlc
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# no shutdown
```

HDLC (High-Level Data Link Control) is Cisco's default serial encapsulation and is deliberately minimal — its job is only to frame data for transmission across the link (mark where a frame starts/ends, provide basic error detection via a frame check sequence).

It has no field or handshake defined for exchanging a username, password, or any credential; it simply assumes that whatever is connected to the serial interface is trusted. What's missing, in security terms, is any identity-verification step: HDLC will bring the link up and pass traffic as soon as the two ends can frame data to each other correctly, with no check on who or what is on the other end.

Task 3: Set up a PPP connection between two routers in Packet Tracer and enable PAP authentication. Show the configuration commands and explain what happens if the password is incorrect.

Router R1:

```
R1(config)# interface serial 0/0/0
R1(config-if)# encapsulation ppp
R1(config-if)# ppp authentication pap
R1(config-if)# ppp pap sent-username baroda password Cisco123
R1(config)#username surat password cisco456
```

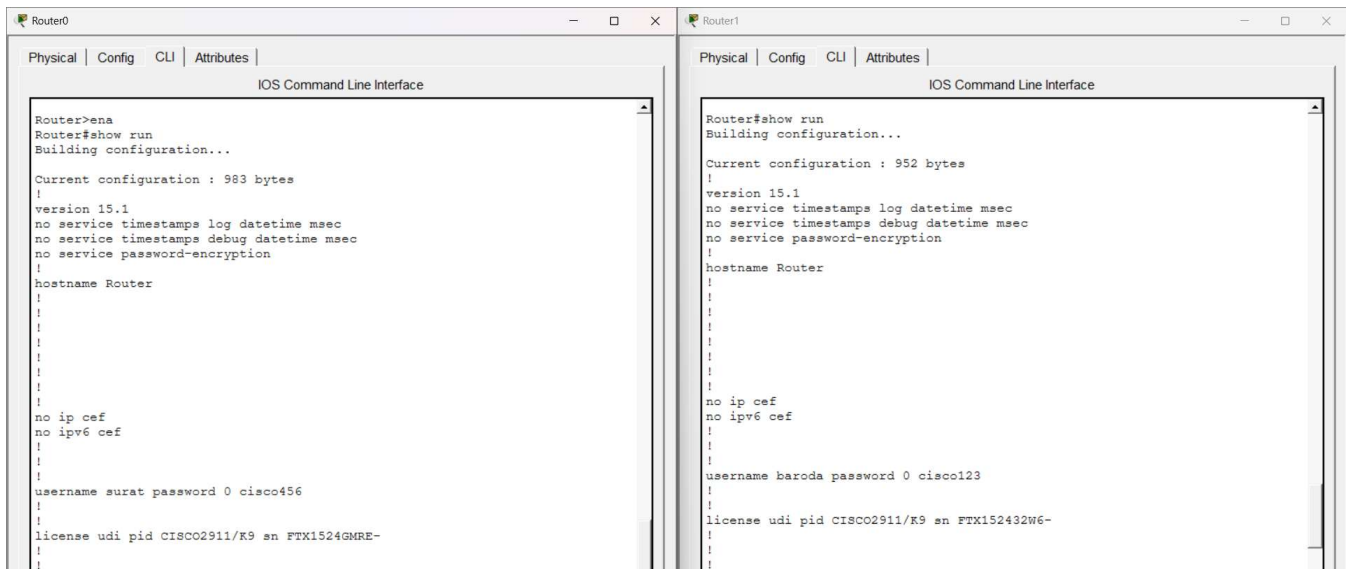
Router R2:

```
R2(config)# interface serial 0/0/0
R2(config-if)# encapsulation ppp
R2(config-if)# ppp authentication pap
R2(config-if)# ppp pap sent-username surat password Cisco456
R2(config)#username baroda password cisco123
```

PAP works as a simple two-way handshake: once the link comes up, the requesting router sends its username and password to the peer in clear text, and the peer checks that pair against what it expects (configured via the sent-username command on each side, matched against the other's ppp pap credentials). If the password is incorrect,

authentication fails, PPP negotiation does not complete, and the line protocol on the serial interface stays down (show interfaces would report "LCP not opened" or similar) — no IP traffic passes over the link until the correct credentials are configured and the interfaces are re-negotiated.

- **ppp pap sent-username** baroda password cisco123 on Router0 = "when I (Router0) connect, here's the credential I will *send out* to prove who I am."
- **username surat password cisco456** on Router0 = "here's what I *expect the peer* to send me, so I can check it."



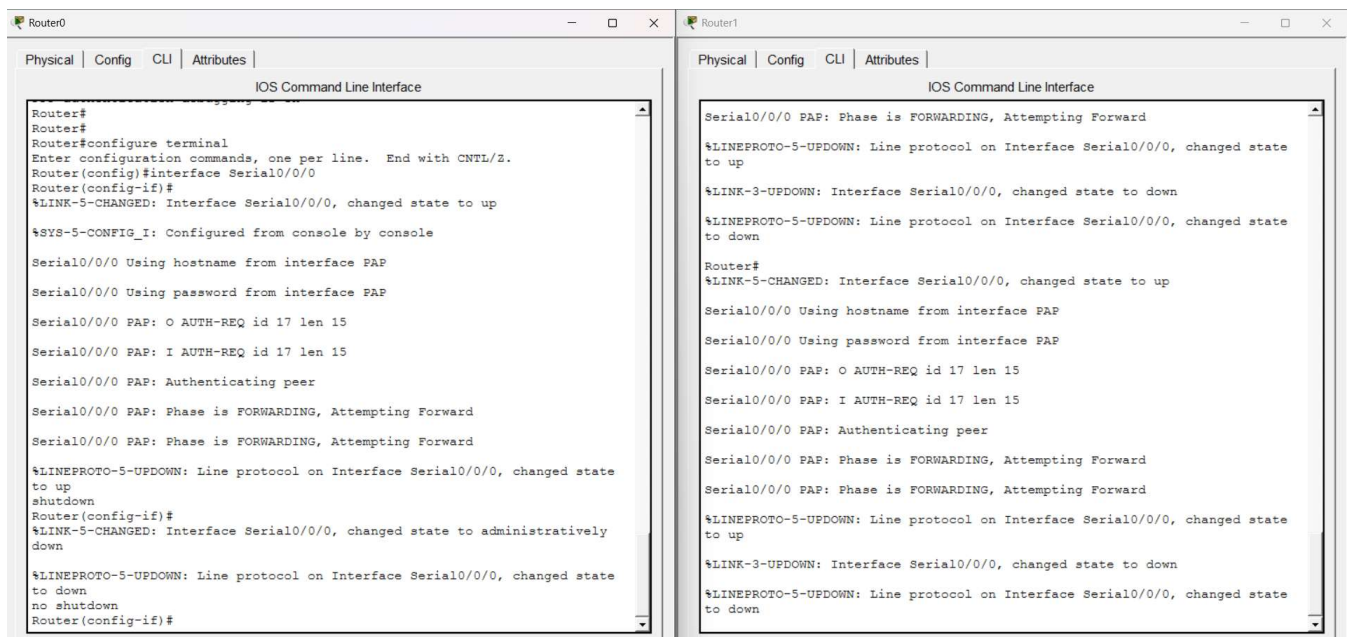
The image shows two side-by-side screenshots of Cisco IOS Command Line Interface (CLI) windows. The left window is for Router0 and the right window is for Router1. Both windows show the output of the 'show run' command, displaying the current configuration of the routers. Router0's configuration includes version 15.1, no service timestamps, no service password-encryption, hostname Router, no ip cef, no ipv6 cef, username surat password 0 cisco456, and license udi pid C1SCO2911/K9 sn FTX1524GMRE-. Router1's configuration includes version 15.1, no service timestamps, no service password-encryption, hostname Router, no ip cef, no ipv6 cef, username baroda password 0 cisco123, and license udi pid C1SCO2911/K9 sn FTX152432W6-.

```
Router0>ena
Router#show run
Building configuration...

Current configuration : 983 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username surat password 0 cisco456
!
!
license udi pid C1SCO2911/K9 sn FTX1524GMRE-
!
!

Router1#show run
Building configuration...

Current configuration : 952 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username baroda password 0 cisco123
!
!
!
license udi pid C1SCO2911/K9 sn FTX152432W6-
!
!
!
```



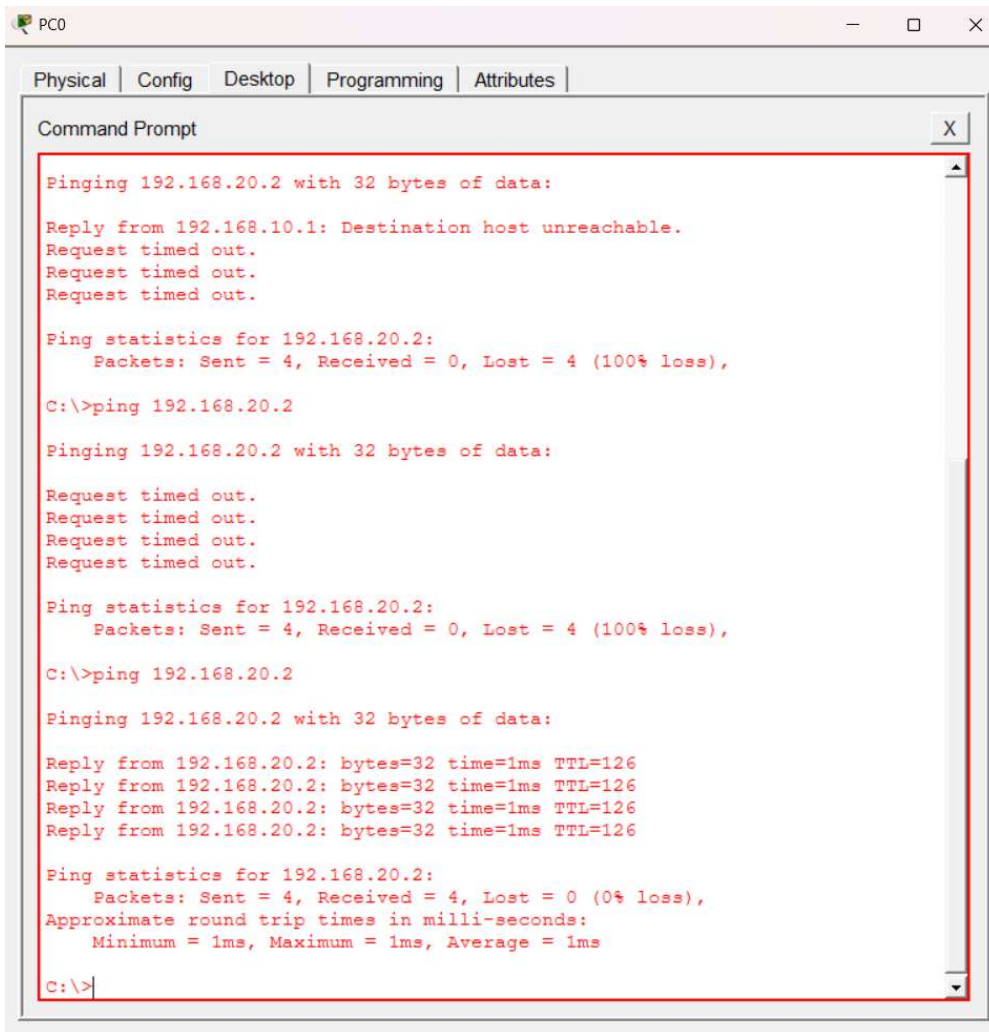
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```
Router0#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/0/0
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%SYS-5-CONFIG_I: Configured from console by console
Serial0/0/0 Using hostname from interface PAP
Serial0/0/0 Using password from interface PAP
Serial0/0/0 PAP: O AUTH-REQ id 17 len 15
Serial0/0/0 PAP: I AUTH-REQ id 17 len 15
Serial0/0/0 PAP: Authenticating peer
Serial0/0/0 PAP: Phase is FORWARDING, Attempting Forward
Serial0/0/0 PAP: Phase is FORWARDING, Attempting Forward
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state
to up
shut down
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to administratively
down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state
to down
no shut down
Router(config-if)#

Router1#
Serial0/0/0 PAP: Phase is FORWARDING, Attempting Forward
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state
to up
%LINK-3-UPDOWN: Interface Serial0/0/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state
to down
Router#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
Serial0/0/0 Using hostname from interface PAP
Serial0/0/0 Using password from interface PAP
Serial0/0/0 PAP: O AUTH-REQ id 17 len 15
Serial0/0/0 PAP: I AUTH-REQ id 17 len 15
Serial0/0/0 PAP: Authenticating peer
Serial0/0/0 PAP: Phase is FORWARDING, Attempting Forward
Serial0/0/0 PAP: Phase is FORWARDING, Attempting Forward
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state
to up
%LINK-3-UPDOWN: Interface Serial0/0/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state
to down
```

After altering the password in R0 – retried the handshake by shutting down and restarting the serial port – Authentication failed warning received

After configuring the correct password, the serial port was shutdown and restarted. Authentication successful and ping worked.



The screenshot shows a window titled "PC0" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the results of three ping attempts to the IP address 192.168.20.2. The first two attempts fail with "Destination host unreachable" and "Request timed out" messages, indicating a 100% loss of packets. The third attempt is successful, showing replies from 192.168.20.2 with 32 bytes of data, a time of 1ms, and a TTL of 126, resulting in a 0% loss of packets.

```
Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=1ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

Ping failed with
wrong password
authentication

Ping success with
correct password
authentication

Task 4: Modify your PPP configuration to use CHAP instead of PAP, and describe one key difference in how CHAP improves security over PAP.

Router R1:

```
R1(config)# username R2 password Cisco123
R1(config)# interface serial 0/0/0
R1(config-if)# encapsulation ppp
R1(config-if)# ppp authentication chap
R1(config-if)# ppp pap sent-username baroda password Cisco123
R2(config)# username surat password cisco123
```

Router R2:

```
R2(config)# username R1 password Cisco123
R2(config)# interface serial 0/0/0
R2(config-if)# encapsulation ppp
R2(config-if)# ppp authentication chap
R2(config-if)# ppp pap sent-username surat password Cisco123
R2(config)# username baroda password cisco123
```

Key difference: CHAP never sends the password itself across the link. Instead it uses a three-way challenge-response handshake — the password here needs to be the same unlike in PAP.

The router sends a challenge, the peer combines it with the shared secret and runs it through a one-way hash (MD5), and returns only the hash. The password itself is never transmitted, so it cannot be captured by sniffing the link the way a PAP exchange (which sends the password in clear text) can. CHAP also re-issues challenges periodically during the session, whereas PAP only authenticates once when the link first comes up.

Task 5: Research and list two real-world scenarios where WAN authentication protocols like PAP, CHAP, or MS-CHAP are used, and explain why authentication is critical in each case.

1. Home/office broadband via PPPoE

Most DSL and fibre broadband connections use PPPoE (PPP over Ethernet) between the customer's router/modem and the ISP, authenticated with PAP, CHAP, or MS-CHAP using the subscriber's account credentials. This is critical because it is how the ISP confirms the connection belongs to a paying, registered subscriber before granting access to the internet — without it, anyone who could physically tap into the line could get free, unbilled, and untraceable internet access.

2. Enterprise leased-line/MPLS access to a service provider's network

When a business router connects to a telecom provider's WAN edge for a leased line or MPLS circuit, PPP with CHAP is commonly used to authenticate the customer's router before it is allowed to send traffic into the provider's backbone. This matters because the provider's backbone carries traffic for many different customers — authentication ensures only the legitimate, provisioned customer router can inject traffic onto that circuit, preventing an unauthorized device from accessing another customer's private routes or consuming bandwidth that isn't theirs.